where $\Delta\tau_1, \Delta\tau_2, \ldots, \Delta\tau_{r-1}$ are the times between collisions, $\Delta\tau_0$ is the time until the first collision, and $\Delta\tau_r$ is the time between the last collision and τ. Here $\mathscr{G}_t$ is the flow map of the smooth system (Hamiltonian H). The trajectory is thus piecewise smooth with continuous configurational path and momenta exhibiting finite jump discontinuities.

Various numerical methods have been proposed for collisional Hamiltonian systems [136, 176, 184, 263, 348, 352]. Typically, these schemes rely on the Verlet method to propagate the system between collisions, with collisions detected either (i) by checking for overlap at the end of the step, (ii) checking for overlap during the step, or (iii) approximating the time to collision before the step. Collisions lead to momentum exchange between particles according the principle outlined above.

3.7.1 Splitting Methods for Hard Sphere Molecular Dynamics

Splitting methods are suggested by considering a formal hard-sphere Hamiltonian

$$H_{\text{h.s.}} = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q}) + U_{\text{h.s.}},$$

where $U_{\text{h.s.}}$ is assumed to be infinite for overlapping configurations (some $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| \leq \sigma_i + \sigma_j$) and zero otherwise. The hard sphere system described by

$$H_{\text{free}} = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U_{\text{h.s.}}$$

consists of purely ballistic (straight line) motion punctuated by momentum jumps. This system is formally integrable, i.e. we may write down its exact solution for any given initial conditions. (In practice, for large systems solved in parallel, this computation may be highly nontrivial [108, 336].) One approach is to consider the splitting $H = H_{\text{free}} + U$, evolving H_{free} for fixed intervals punctuated by impulses derived from the smooth potential U. In [40, 184] this algorithm is termed the "Primitive Splitting Algorithm" and can be described by the three steps:

$$\boldsymbol{P} := \boldsymbol{p} - \frac{h}{2}\nabla U(\boldsymbol{q}), \qquad (\boldsymbol{Q}, \boldsymbol{P}) := \mathscr{G}_h^{\text{free}}(\boldsymbol{q}, \boldsymbol{P}), \qquad \boldsymbol{P} := \boldsymbol{P} - \frac{h}{2}\nabla U(\boldsymbol{Q}).$$

Even with this symmetric form ("kick", "drift", "kick") where "drift" now involves the solution of the system H_{free}, it was shown in [184] that energy accumulates rapidly. Assuming a finite number of collisions on a fixed interval, the error behaves as $\mathcal{O}(h)$. In long simulations the energy error grows without bound.

To understand why this is the case, we consider a step of the primitive splitting algorithm applied to a simple system consisting of a fixed body at the origin and a single body (mass=1) moving under an external potential and in occasional contact with the fixed body, both bodies having radius 0.5. Let us consider three possibilities: (i) there is no collision in the timestep, (ii) there is a collision only at